1 **(a)** Uses Pythagoras' theorem: $AC^2 = 8^2 + 6^2$ **[M1]**

$AC = 10$ **[A1]**

(b) Uses 3D Pythagoras' theorem: $AG^2 = 10^2 + 5^2$ (or $8^2 + 6^2 + 5^2$) **[M1]**

$AG = \sqrt{125}$ **[A1]**

$5\sqrt{5}$ **[A1]**

[Total Marks: 5]

2 **(a)** $BF^2 = 6^2 + 8^2 = 36 + 64 = 100$ **[M1]**

10 **[A1]**

(b) $AF^2 = (10\sqrt{3})^2 + 10^2 = 300 + 100 = 400$ **[M1]**

20 **[A1]**

(c) $\tan \angle AFB = \dfrac{10\sqrt{3}}{10} = \sqrt{3}$ (or $\sin \angle AFB = \dfrac{10\sqrt{3}}{20} = \dfrac{\sqrt{3}}{2}$) **[M1]**

60 **[A1]**

(d) $AC^2 = (10\sqrt{3})^2 + 6^2 = 300 + 36 = 336$ **[M1]**

$AC = \sqrt{336} = 4\sqrt{21}$ cm **[A1]**

Area $= 4\sqrt{21} \times 8 = 32\sqrt{21}$ **[A1]**

[Total Marks: 9]

3 **(a)** $AC^2 = 8^2 + 6^2$ **[M1]**

$AC = 10$ **[A1]**

(b) $AG^2 = 10^2 + 5^2$ (or $AG^2 = 8^2 + 6^2 + 5^2$) **[M1]**

$AG = \sqrt{125} = 11.18...$ so 11.2 **[A1]**

(c) $\tan \theta = \dfrac{5}{10}$ (or $\sin \theta = \dfrac{5}{\sqrt{125}}$, etc.) **[M1]**

$\theta = 26.56...$ so $26.6°$ **[A1]**

[Total Marks: 6]

4 **(a)** $AC^2 = 12^2 + 5^2 = 144 + 25 = 169$ **[M1]**

13 **[A1]**

(b) $AG^2 = 13^2 + 8^2 = 169 + 64 = 233$ **[M1]**

15.3 **[A1]**

(c) Angle between AG and base $ABCD$ is angle GAC **[B1]**

$\tan \angle GAC = \dfrac{8}{13}$ **[M1]**

31.6 **[A1]**

[Total Marks: 7]

5 **(a)** Uses Pythagoras to find half diagonal AM: $AM = \frac{1}{2}\sqrt{10^2 + 10^2} = \sqrt{50}$ (or 7.07...) **[M1]**

$VA^2 = 12^2 + (\sqrt{50})^2 = 144 + 50 = 194$ **[M1]**

$VA = \sqrt{194} = 13.92...$ so 13.9 **[A1]**

(b) $\tan(\angle VAM) = \frac{12}{\sqrt{50}}$ **[M1]**

$\angle VAM = 59.49...$ so $59.5°$ **[A1]**

[Total Marks: 5]

6 **(a)** $BD^2 = 14^2 + 8^2 = 196 + 64 = 260$ **[M1]**

16.1 (or 16.12...) **[A1]**

(b) Identifies right-angled triangle FBD with the right angle at B (implied by using tan) **[M1]**

$\tan(\angle FDB) = \frac{6}{\sqrt{260}}$ (or $\frac{6}{16.12...}$) **[M1]**

20.4 (or 20.41 to 20.42) **[A1]**

(c) Identifies the horizontal distances from A to M as 7 cm and 8 cm, and vertical distance as 6 cm **[M1]**

Uses 3D Pythagoras' theorem: $AM^2 = 7^2 + 8^2 + 6^2$ **[M1]**

12.2 (or 12.20...) **[A1]**

[Total Marks: 8]

7 **(a)** Uses Pythagoras to find AC: $AC = \sqrt{12^2 + 5^2} = 13$ **[M1]**

$AF^2 = 13^2 + 8^2 = 169 + 64 = 233$ **[M1]**

$AF = \sqrt{233} = 15.26...$ so 15.3 **[A1]**

(b) Identifies required angle as $\angle FAC$ and uses trig, e.g. $\tan(\angle FAC) = \frac{8}{13}$ **[M1]**

$\angle FAC = 31.60...$ so $31.6°$ **[A1]**

(c) Recognises the required angle is $\angle FAE$ because FE is perpendicular to plane $ABED$ **[B1]**

Uses trig in triangle FAE (where $\angle FEA = 90°$): $\sin(\angle FAE) = \frac{5}{\sqrt{233}}$ OR finds $AE = \sqrt{12^2 + 8^2} = \sqrt{208}$ and uses $\tan(\angle FAE) = \frac{5}{\sqrt{208}}$ **[M1]**

$\angle FAE = 19.11...$ so $19.1°$ **[A1]**

[Total Marks: 8]

8 **(a)** Uses Volume $= \frac{1}{3} \times$ base area $\times$ height $\implies 384 = \frac{1}{3} \times 12^2 \times VO$ **[M1]**

$48 \times VO = 384 \implies VO = 8$ **[A1]**

(b) Calculates $AO = \frac{1}{2}\sqrt{12^2 + 12^2} = 6\sqrt{2}$ (or 8.48...) **[M1]**

Uses Pythagoras' theorem in triangle VOA: $VA^2 = 8^2 + (6\sqrt{2})^2 = 64 + 72 = 136$ **[M1]**

11.7 (or 11.66...) **[A1]**

(c) $\tan(\angle VAO) = \frac{8}{6\sqrt{2}}$ (or $\frac{8}{8.485}$) **[M1]**

43.3 (or 43.31 to 43.32) **[A1]**

(d) Identifies the distance from O to the midpoint of AB is 6 cm **[B1]**

Let M be the midpoint of AB. $\tan(\angle VMO) = \frac{8}{6}$ **[M1]**

53.1 (or 53.13 to 53.14) **[A1]**

[Total Marks: 10]

9 **(a)** $AC = \sqrt{16^2 + 12^2} = \sqrt{400} = 20$ cm, giving $OA = 10$ cm **[M1]**

$VO^2 = 26^2 - 10^2 = 676 - 100 = 576$ **[M1]**

$VO = \sqrt{576} = 24$ cm **[A1]**

(b) $\tan \angle VAO = \frac{24}{10}$ (or $\sin \angle VAO = \frac{24}{26}$) **[M1]**

67.4 **[A1]**

(c) **(i)** Angle required is angle VMO, where $OM = \frac{16}{2} = 8$ cm **[B1]**

$\tan \angle VMO = \frac{24}{8} = 3$ **[M1]**

71.6 **[A1]**

(ii) $VM = \sqrt{24^2 + 8^2} = \sqrt{640}$ $(= 25.298\ldots)$ cm **[M1]**

Shortest distance $d = 8 \sin 71.565\ldots^\circ$ (or $\frac{24 \times 8}{\sqrt{640}} = 7.589\ldots$) **[M1]**

7.59 **[A1]**

[Total Marks: 11]